Complex Analysis Homework 7

1. Show that $g_k(x) = \frac{x^k}{1+x^k}$ converges pointwisely on $[0, \infty)$. What is the limit function? Show that $g_k(x)$ does not converges uniformly on any interval I which contains 1. Furthermore, show that $g_k(x)$ converges uniformly on $[r, \infty)$ for any $r > 1$.

Solution

If $0 \leq x < 1$, in this interval we have that $x^k \to 0$ as $k \to \infty$, and we have

$$g(x) = \lim_{k\to\infty} \frac{x^k}{1+x^k} = \frac{0}{1+0} = 0.$$

If $x = 1$, then

$$g(1) = \lim_{k\to\infty} \frac{1^k}{1+1^k} = \frac{1}{1+1} = \frac{1}{2}.$$

If $x > 1$, in this interval we have that $x^k \to 0$ as $k \to \infty$, and we have

$$g(x) = \lim_{k\to\infty} \frac{x^k}{1+x^k} = \lim_{k\to\infty} \frac{1}{1/x^k+1} = \frac{1}{0+1} = 1.$$

We also have that $g_k(x) \to g(x)$, where

$$g(x) = \begin{cases} 0 & \text{for} \quad 0 \leq x < 1, \\ \frac{1}{2} & \text{for} \quad x = 1, \\ 1 & \text{for} \quad x > 1 \end{cases}$$

$g_k(x)$ does not converges uniformly on any interval I which contains 1 because limit function is not continuous at 1.

$$\left|\frac{x^k}{1+x^k} - 1\right| = \frac{1}{1+x^k} \leq \frac{1}{1+r^k} < \varepsilon \quad \Rightarrow \quad r^k > \frac{1}{\varepsilon} - 1 \quad \Rightarrow \quad k > \ln\left(\frac{1}{\varepsilon} - 1\right) / \ln r$$

Choose $N = \left\lfloor \ln\left(\frac{1}{\varepsilon} - 1\right) / \ln r \right\rfloor + 1$ so that $N > \ln\left(\frac{1}{\varepsilon} - 1\right) / \ln r.$

For any $\varepsilon > 0,$ there exists an positive integer $N = \left\lfloor \ln\left(\frac{1}{\varepsilon} - 1\right) / \ln r \right\rfloor + 1$ such that

$$\left| \frac{x^k}{1+x^k} - 1 \right| = \frac{1}{1+x^k} \le \frac{1}{1+r^k} < \varepsilon$$

for all $x \ge r,$ for all $k \ge N.$

2. (a) Verify that the identity $\text{Log}\left(\frac{1+z}{1-z}\right) = \text{Log}(1+z) - \text{Log}(1-z)$ holds when $|z| < 1$. Then, using the Maclaurin expansions of $\text{Log}(1 + z)$ and $\text{Log}(1-z)$, find the Maclaurin expansion of $\text{Log}\left(\frac{1+z}{1-z}\right)$.

(b) Find the power series expansion of the principal branch $\text{Tan}^{-1}(z)$ of the inverse tangent function about $z = 0$. What is the radius of convergence of the series?

(c) Expand $\text{Log}(1 + iz)$ and $\text{Log}(1-iz)$ in power series about $z = 0$. By comparing power series expansions, establish the identity $\text{Tan}^{-1}(z) = \frac{1}{2i}\text{Log}\left(\frac{1+iz}{1-iz}\right)$ when $|z| < 1$.

Solution

(a) $$\text{Log}\left(\frac{1+z}{1-z}\right) = \text{Log}\left|\frac{1+z}{1-z}\right| + i\text{Arg}\left(\frac{1+z}{1-z}\right)$$
$$= \text{Log}|1+z| - i\text{Log}|1-z| + i\text{Arg}\left(\frac{1+z}{1-z}\right)$$

Now $|z| < 1$ insures that $1 + z$ and $1 - z$ lie in the right half plane so this equals

$$\text{Log}|1+z| - \text{Log}|1-z| + i\text{Arg}(1+z) - i\text{Arg}(1-z)$$
$$= \text{Log}(1+z) - \text{Log}(1-z)$$

It follows that $$\text{Log}\left(\frac{1+z}{1-z}\right) = \sum_{j=1}^{\infty} \frac{(-1)^{j+1} z^j}{j} - \sum_{j=1}^{\infty} \frac{-z^j}{j}$$
$$= 2\sum_{k=0}^{\infty} \frac{z^{2k+1}}{2k+1}$$

(b) $\frac{d}{dz}\mathrm{Tan}^{-1}(z)=\frac{1}{1+z^2}=\sum_{k=0}^{\infty}(-z^2)^k \quad |z|<1$

$\sum_{k=0}^{\infty}(-z^2)^k$ converges uniformly in any closed subdisk $|z|\le R'<1$ by Theorem 7 in P.253

By Theorem 8 in P.255, $\int_0^z \frac{d}{d\omega}\mathrm{Tan}^{-1}(\omega)=\sum_{k=0}^{\infty}\int_0^z(-1)^k\omega^{2k}d\omega$

$\mathrm{Tan}^{-1}(z)=\sum_{k=0}^{\infty}(-1)^k\frac{z^{2k+1}}{2k+1}$

$R=\lim_{k\to\infty}\left|\frac{2k+3}{2k+1}\right|=1$

(c) $\mathrm{Log}(1+iz)=-\sum_{k=0}^{\infty}\frac{(-iz)^{k+1}}{k+1}, \quad \mathrm{Log}(1-iz)=-\sum_{k=0}^{\infty}\frac{(iz)^{k+1}}{k+1}, \quad |z|<1.$

$$\frac{1}{2i}\mathrm{Log}\left(\frac{1+iz}{1-iz}\right)=\frac{1}{2i}\left[-\sum_{k=0}^{\infty}\frac{(-iz)^{k+1}}{k+1}+\sum_{k=0}^{\infty}\frac{(iz)^{k+1}}{k+1}\right]$$
$$=\frac{1}{2i}\sum_{k=0}^{\infty}2\frac{(iz)^{2k+1}}{2k+1}$$
$$=\sum_{k=0}^{\infty}i^{2k}\frac{z^{2k+1}}{2k+1}$$
$$=\sum_{k=0}^{\infty}(-1)^k\frac{z^{2k+1}}{2k+1}$$
$$=\mathrm{Tan}^{-1}(z)$$

3. Express the following function in power series at z_0 and find their radii of convergence.

(a) $\frac{e^z}{1+z}$ at 0

(b) $\frac{z^2-1}{z^4-1}$ at 2

Solution

(a) We have

$$\frac{1}{1+z}=\sum_{k=0}^{\infty}(-1)^k z^k \quad \text{for all } |z|<1 \qquad \text{and} \qquad \sum_{k=0}^{\infty}\frac{z^k}{k!} \quad \text{for all } z\in\mathbb{C}.$$

Since the given function can be expressed as product of these two we have

$$\frac{e^z}{1+z}=\frac{1}{1+z}e^z=\sum_{n=0}^{\infty}a_n z^n \qquad \text{for all } |z|<1$$

where

$$\begin{aligned} a_n &= \frac{(-1)^n}{0!}+\frac{(-1)^{n-1}}{1!}+\frac{(-1)^{n-2}}{2!}+\cdots+\frac{(-1)^0}{n!} \\ &= (-1)^n\left[\frac{1}{2!}-\frac{1}{3!}+\cdots+\frac{(-1)^n}{n!}\right]. \end{aligned}$$

$$R=\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|=\lim_{n\to\infty}\left|\frac{\frac{1}{2!}-\frac{1}{3!}+\cdots+\frac{(-1)^n}{n!}}{\frac{1}{2!}-\frac{1}{3!}+\cdots+\frac{(-1)^n}{n!}+\frac{(-1)^{n+1}}{(n+1)!}}\right|=\lim_{n\to\infty}\left|1-\frac{\frac{(-1)^{n+1}}{(n+1)!}}{\frac{1}{2!}-\frac{1}{3!}+\cdots+\frac{(-1)^n}{n!}+\frac{(-1)^{n+1}}{(n+1)!}}\right|=1$$

(b) $$\begin{aligned} \frac{z^2-1}{z^4-1} &= \frac{1}{(z-i)(z+i)}=\frac{0.5i}{z+i}-\frac{0.5i}{z-i} \\ &= \frac{0.5i}{z-2+2+i}-\frac{0.5i}{z-2+2-i} \\ &= \frac{0.5i}{2+i}\frac{1}{1+\frac{z-2}{2+i}}-\frac{0.5i}{2-i}\frac{1}{1+\frac{z-2}{2-i}} \qquad \left|\frac{z-2}{2-i}\right|=\left|\frac{z-2}{2+i}\right|<1 \\ &= \frac{0.5i}{2+i}\left(1-\tfrac{z-2}{2+i}+(\tfrac{z-2}{2+i})^2-(\tfrac{z-2}{2+i})^3+\cdots\right)-\frac{0.5i}{2-i}\left(1-\tfrac{z-2}{2-i}+(\tfrac{z-2}{2-i})^2-(\tfrac{z-2}{2-i})^3+\cdots\right) \\ &= \sum_{j=0}^{\infty}a_j z^j, \qquad a_j=(-1)^j\left(\frac{0.5i}{(2+i)^{j+1}}-\frac{0.5i}{(2-i)^{j+1}}\right) \end{aligned}$$

$R=|2+i|=|2-i|=\sqrt{5}$

4 (a) Show that $f(z)=\int_0^1 \frac{\sin zt}{t}dt$ is an entire function by obtaining a power series expansion for f.

(b) Show that $f'(z)=\int_0^1 \cos ztdt$ by using the power series for f.

Solution

(a) $f(z)=\int_0^1 \frac{\sin zt}{t}dt$

$$=\int_0^1 \frac{1}{t}\left(zt-\frac{(zt)^3}{3!}+\frac{(zt)^5}{5!}-\frac{(zt)^7}{7!}+\cdots\right)dt$$

$$=\int_0^1 \left(z-\frac{z^3}{3!}t^2+\frac{z^5}{5!}t^4-\frac{z^7}{7!}t^6+\cdots\right)dt$$

$$=z-\frac{z^3}{3(3!)}+\frac{z^5}{5(5!)}-\frac{z^7}{7(7!)}+\cdots$$

$$R=\lim_{j\to\infty}\left|\frac{a_j}{a_{j+1}}\right|=\lim_{j\to\infty}\left|\frac{(2j+1)\cdot((2j+1)!)}{(2j-1)\cdot((2j-1)!)}\right|=\lim_{j\to\infty}\left|\frac{(2j+1)^2(2j)}{(2j-1)}\right|=\infty$$

(b) $\int_0^1 \cos ztdt=\int_0^1\left(1-\frac{(zt)^2}{2!}+\frac{(zt)^4}{4!}-\frac{(zt)^6}{6!}+\cdots\right)dt=1-\frac{z^2}{3!}+\frac{z^4}{5!}-\frac{z^6}{7!}+\cdots=f'(z)$

5. Consider the following infinite series of complex functions $\sum_{k=1}^{\infty}\frac{\sin kz}{k^2}$. Show that it is absolutely convergent when z is real, but that when z is non-real, it becomes divergent.

Solution

(i) When z is real, we have

$$\left|\frac{\sin kz}{k^2}\right| \leq \frac{1}{k^2}, \qquad \text{for all positive integer values of } k.$$

Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is known to be convergent, $\sum_{k=1}^{\infty} \frac{\sin kz}{k^2}$ is absolutely convergent.

(ii) When z is non-real, we let $z = x + iy, \ y \neq 0$. From the relation

$$\frac{\sin kz}{k^2} = \frac{e^{-ky}e^{ikx} - e^{ky}e^{-ikx}}{2k^2 i},$$

we deduce that

$$\left|\frac{\sin kz}{k^2}\right| \geq \frac{e^{k|y|} - e^{-k|y|}}{2k^2} \to \infty \text{ as } k \to \infty.$$

Since $\left|\frac{\sin kz}{k^2}\right|$ is unbounded as $k \to \infty$, $\sum_{k=1}^{\infty} \frac{\sin kz}{k^2}$ is divergent.